Ch02: Convex Sets

Convex Optimization Book by Lieven Vandenberghe and Stephen P. Boyd

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1 Operations that preserve convexity

Exercise 2.16

Show that if S_1 and S_2 are convex sets in $\mathbb{R}^{m\times n}$, then so is their partial sum

$$S = \{(x,\, y_1 + y_2) \mid x \in \mathbb{R}^m,\ y_1, y_2 \in \mathbb{R}^n,\ (x, y_1) \in S_1,\ (x, y_2) \in S_2\}.$$

Answer:

Before proving 2.16, we first prove a few claims and then return to the main problem.

Claim 1: Image of a Convex Set under an Affine Function is Convex

Claim. If $S \subseteq \mathbb{R}^n$ is convex, then the image of S under an affine function is also convex.

Proof. Let $S \subseteq \mathbb{R}^n$ be convex and consider

$$f(x) = Ax + b, \qquad A \in \mathbb{R}^{m\times n},\ b \in \mathbb{R}^m.$$

We want to show that $f(S)$ is convex.

Take any $x_1, x_2 \in S$ and $\theta \in [0,1]$. Since S is convex,

$$\theta x_1 + (1-\theta)x_2 \in S.$$

Then

$$\begin{aligned} f(\theta x_1 + (1-\theta)x_2) &= A(\theta x_1 + (1-\theta)x_2) + b \\ &= \theta(Ax_1 + b) + (1-\theta)(Ax_2 + b) \\ &= \theta f(x_1) + (1-\theta) f(x_2). \end{aligned}$$

Hence, $\theta f(x_1) + (1-\theta) f(x_2) \in f(S)$, so $f(S)$ is convex.

□

Claim 2: Sum of Two Convex Sets is Convex

Claim. If $S_1, S_2 \subseteq \mathbb{R}^m$ are convex, then their sum

$$S_1 + S_2 = \{\, x_1 + x_2 \mid x_1 \in S_1,\ x_2 \in S_2 \,\}$$

is convex.

Proof. Let $x_1, x_1' \in S_1$ and $x_2, x_2' \in S_2$. Then $x_1 + x_2$, $x_1' + x_2' \in S_1 + S_2$. We need to show that for any $\theta \in [0, 1]$,

$$\theta(x_1 + x_2) + (1 - \theta)(x_1' + x_2') \in S_1 + S_2.$$

We have

$$\theta(x_1 + x_2) + (1 - \theta)(x_1' + x_2') = [\theta x_1 + (1 - \theta)x_1'] + [\theta x_2 + (1 - \theta)x_2'].$$

Since S_1 and S_2 are convex,

$$\theta x_1 + (1 - \theta)x_1' \in S_1, \qquad \theta x_2 + (1 - \theta)x_2' \in S_2,$$

so their sum belongs to $S_1 + S_2$. Hence $S_1 + S_2$ is convex.

$\square$

Now Coming to Problem 2.16

Let $S_1, S_2 \subseteq \mathbb{R}^{m+n}$ be convex sets. Here $x \in \mathbb{R}^m$ and $y_1, y_2 \in \mathbb{R}^n$.

Step 1: Affine transformations for each component. We define the following affine functions to extract each component. Each affine function maps from $\mathbb{R}^{m+n}$ to its respective subspace.

For x, we define

$$x = \begin{bmatrix} I_{m\times m} & 0_{n\times n} \end{bmatrix} \begin{bmatrix} x \\ y_1 \end{bmatrix}, \qquad \text{where } \begin{bmatrix} x \\ y_1 \end{bmatrix} \in S_1 \subset \mathbb{R}^{m+n}.$$

For y_1, we define

$$y_1 = \begin{bmatrix} 0_{m\times m} & I_{n\times n} \end{bmatrix} \begin{bmatrix} x \\ y_1 \end{bmatrix}, \qquad \text{where } \begin{bmatrix} x \\ y_1 \end{bmatrix} \in S_1 \subset \mathbb{R}^{m+n}.$$

Similarly, for y_2, we define

$$y_2 = \begin{bmatrix} 0_{m\times m} & I_{n\times n} \end{bmatrix} \begin{bmatrix} x \\ y_2 \end{bmatrix}, \qquad \text{where } \begin{bmatrix} x \\ y_2 \end{bmatrix} \in S_2 \subset \mathbb{R}^{m+n}.$$

Under affine transformations of convex sets, the images are convex. Hence, x, y_1, and y_2 are convex sets individually.

Step 2: Sum of convex sets. Since $y_1, y_2 \subseteq \mathbb{R}^n$ are convex sets, their sum

$$y_1 + y_2 = \{\, y_1 + y_2 \mid y_1 \in Y_1,\ y_2 \in Y_2 \,\}$$

is also convex (by Claim 2).

Step 3: Cartesian product of convex sets. Finally, since $x \subseteq \mathbb{R}^m$ and $(y_1 + y_2) \subseteq \mathbb{R}^n$ are convex sets, their Cartesian product

$$\Big\{ (x,\, y_1 + y_2) \mid x \in \mathbb{R}^m,\ y_1, y_2 \in \mathbb{R}^n,\ (x, y_1) \in S_1,\ (x, y_2) \in S_2 \Big\}$$

is convex.

Thus, the set

$$S = \{(x,\, y_1 + y_2) \mid (x, y_1) \in S_1,\ (x, y_2) \in S_2\}$$

is convex.

$\blacksquare$

Exercise 2.17: Image of Polyhedral Sets under Perspective Function

The perspective function is

$$P(x,t) = \frac{x}{t}, \qquad \operatorname{dom} P = \mathbb{R}^n \times \mathbb{R}_{++}.$$

For each set C, we describe

$$P(C) = \left\{ \frac{v}{t} \;\middle|\; (v,t) \in C,\ t > 0 \right\}.$$

(a) The Polyhedron

Let

$$C = \operatorname{conv}\{(v_1, t_1), (v_2, t_2), \ldots, (v_K, t_K)\},$$

where $v_i \in \mathbb{R}^n$ and $t_i > 0$.

The set C is the convex hull of a finite number of points:

$$C = \left\{ \sum_{i=1}^{K} \theta_i \begin{bmatrix} v_i \\ t_i \end{bmatrix} \;\middle|\; \theta_i \geq 0,\ \sum_{i=1}^{K} \theta_i = 1 \right\}.$$

Thus, any point $(v,t) \in C$ can be expressed as

$$(v,t) = \sum_{i=1}^{K} \theta_i (v_i, t_i) = \left(\sum_{i=1}^{K} \theta_i v_i,\ \sum_{i=1}^{K} \theta_i t_i \right), \qquad \text{where } \theta_i \geq 0,\ \sum_{i=1}^{K} \theta_i = 1.$$

Applying the perspective function $P(x,t) = x/t$, we have

$$P(C) = \left\{ \frac{\sum_{i=1}^{K} \theta_i v_i}{\sum_{i=1}^{K} \theta_i t_i} \;\middle|\; \theta_i \geq 0,\ \sum_{i=1}^{K} \theta_i = 1 \right\}.$$

Define normalized weights

$$\alpha_i = \frac{\theta_i t_i}{\sum_{j=1}^{K} \theta_j t_j}, \qquad \alpha_i \geq 0,\ \sum_{i=1}^{K} \alpha_i = 1.$$

Then

$$\frac{\sum_{i=1}^K \theta_i v_i}{\sum_{i=1}^K \theta_i t_i} = \sum_{i=1}^K \alpha_i \frac{v_i}{t_i}.$$

Hence, the image of C under the perspective map is the convex hull of the scaled points $\frac{v_i}{t_i}$:

$$P(C) = \operatorname{conv}\Big\{\frac{v_1}{t_1}, \frac{v_2}{t_2}, \dots, \frac{v_K}{t_K}\Big\}.$$

$\blacksquare$

(b) The Hyperplane

Let

$$C = \{\, (v,t) \mid f^T v + gt = h \,\}, \qquad (f,g) \neq (0,0).$$

Apply the perspective map by dividing by $t > 0$:

$$f^T \left(\frac{v}{t}\right) + g = \frac{h}{t}.$$

Let $x = v/t$. Then

$$P(C) = \{\, x \mid f^T x + g = h/t,\ t > 0\}.$$

We now analyze all cases depending on h.

Case 1: $h = 0$

Then $h/t = 0$ for all $t > 0$, so

$$f^T x + g = 0.$$

Thus

$$P(C) = \{\, x \mid f^T x + g = 0 \,\},$$

a single hyperplane.

Case 2: $h > 0$

Then

$$\frac{h}{t} \in (0, \infty).$$

As $t \to \infty$, $h/t \to 0^+$; as $t \to 0^+$, $h/t \to +\infty$.

Thus the image consists of infinitely many parallel hyperplanes

$$f^T x + g = \alpha, \qquad \alpha \in [0, \infty),$$

whose union is the halfspace

$$P(C) = \{\, x \mid f^T x + g \geq 0 \,\}.$$

Case 3: $h < 0$

Now

$$\frac{h}{t} \in (-\infty, 0).$$

As t varies, we sweep hyperplanes

$$f^T x + g = \alpha, \qquad \alpha \in (-\infty, 0),$$

whose union is the halfspace

$$P(C) = \{\, x \mid f^T x + g \le 0 \,\}.$$

Intuition. The hyperplane in (v, t) becomes a "moving" hyperplane in x as $1/t$ changes. The union of these parallel hyperplanes fills one side of the limiting hyperplane $f^T x + g = 0$, forming a halfspace (or exactly the hyperplane when $h = 0$).

(c) The Halfspace

Let

$$C = \{\, (v, t) \mid f^T v + gt \le h \,\}.$$

Divide by $t > 0$:

$$f^T x + g \le \frac{h}{t}, \qquad x = v/t.$$

Case 1: $h = 0$

Then

$$f^T x + g \le 0,$$

independent of t. Hence

$$P(C) = \{\, x \mid f^T x + g \le 0 \,\}.$$

Case 2: $h > 0$

Then

$$\frac{h}{t} \in (0, \infty).$$

So the inequality

$$f^T x + g \le \frac{h}{t}$$

sweeps from the halfspace $f^T x + g \le 0$ outward to $f^T x + g \le +\infty$, which eventually includes all of $\mathbb{R}^n$.

Thus

$$P(C) = \mathbb{R}^n.$$

Case 3: $h < 0$

Then

$$\frac{h}{t} \in (-\infty, 0).$$

The boundary never rises above 0, so the union of all halfspaces

$$f^T x + g \le \alpha, \qquad \alpha < 0,$$

is simply the limiting halfspace:

$$P(C) = \{\, x \mid f^T x + g \le 0 \,\}.$$

Intuition. Each t creates a halfspace in x. For $h > 0$ these halfspaces expand until they cover all of $\mathbb{R}^n$. For $h \le 0$, they remain below the limiting hyperplane $f^T x + g = 0$.

(d) The Polyhedron

Let

$$C = \{\, (v, t) \mid Fv + gt \preceq h \,\},$$

with rows (f_i^T, g_i) and h_i.

Applying the perspective map and dividing by $t > 0$ gives

$$f_i^T x + g_i \leq \frac{h_i}{t}, \qquad x = v/t.$$

For each inequality, the behavior matches part (c):

- If $h_i = 0$, then $f_i^T x + g_i \leq 0$.
- If $h_i > 0$, then the corresponding constraint gives all of $\mathbb{R}^n$.
- If $h_i < 0$, then again $f_i^T x + g_i \leq 0$ in the limit.

Thus the final projection is

$$P(C) = \bigcap_{\{i : h_i \leq 0\}} \{\, x \mid f_i^T x + g_i \leq 0 \,\}.$$

Intuition. A polyhedron is an intersection of halfspaces. Under perspective, each inequality either collapses to the limiting halfspace $f_i^T x + g_i \leq 0$ or becomes inactive (when $h_i > 0$). The resulting set is again a polyhedron.